

Infusing Accessibility into Programming Courses

Lin Jia

Rochester Institute of
Technology Rochester, NY, USA
lj7141@g.rit.edu

Catherine M. Baker

Creighton University
Omaha, NE, USA
catherinebaker@creighton.edu

Yasmine N. Elglaly

Western Washington University
Bellingham, WA, USA
elglaly@www.edu

Kristen Shinohara

Rochester Institute of
Technology Rochester, NY, USA
kristen.shinohara@rit.edu

ABSTRACT

Within computing education, accessibility topics are usually taught in Human Computer Interaction and Web Design courses. Few have included accessibility in programming courses as an add-on topic. We studied assignments that infuse accessibility into programming topics without impacting the core computing learning objectives. We present two examples, assignments that can be used in Introductory Programming and Object Oriented Programming courses. Both assignments cover accessibility in addition to the primary computing topic taught. We included the two assignments in two courses for two semesters, conducting surveys and interviews to understand the impact of the assignments on students' learning of accessibility and computing. Our findings show this approach has potential to satisfy accessibility and programming learning objectives without overwhelming the students, though more work is needed to make sure that students are clear on the relationship between the assignments and technical accessibility knowledge.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**; • **Social and professional topics** → **Computing education**.

KEYWORDS

computing education, accessibility

ACM Reference Format:

Lin Jia, Yasmine N. Elglaly, Catherine M. Baker, and Kristen Shinohara. 2021. Infusing Accessibility into Programming Courses. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451625>

1 INTRODUCTION

Accessibility awareness in the technology industry has increased considerably over the years, along with a demand for computing professionals equipped with knowledge and skills to build accessible products. This demand motivated attempts to incorporate

accessibility content into computer science education curricula¹. Yet, many current mainstream technology products do not meet accessibility criteria, indicating gaps in implementing accessibility standards [21]. Practitioners attributed accessibility omission to a lack of relevant knowledge and skills [18, 26]. Indeed, recent studies indicate a paucity of accessibility training in formal education [10, 18]. Thus, we may infer opportunities exist to incorporate accessibility into computing education curricula.

Various approaches to teaching accessibility have been assessed as effective in raising student awareness of accessibility knowledge, yet there are limitations that prevent curricular adoption of accessibility-related topics. Instructors who include accessibility tend to already have expertise in Human-Computer Interaction and related fields [20], suggesting that it may be difficult for instructors unfamiliar with accessibility to incorporate new materials and increase capacity through knowledge and skill-building. Low student engagement and enrollment in elective course offerings promoting accessibility indicate a lack of demand from students and fleeting support from administration [3, 20]. Thus, instructors may need support to integrate accessibility topics into courses [11], and the resources to include accessibility in core computing curricula without disrupting key learning objectives.

We investigated infusing accessibility in the domain or topic of programming assignments while maintaining the integrity of the core learning objectives. Each assignment had at least one core CS learning objective, e.g., learning pointers, and at least one accessibility learning objective, e.g., learning about Braille as an assistive technology used by blind people to access text. We created two assignments to investigate the effectiveness of infusing accessibility. We focused on accessibility knowledge that does not rely on interaction design skills, such as converting ASCII to Braille or researching automated accessibility checkers. Data consisted of student completed pre- and post- surveys, activity journals and reflections on attitudes towards accessibility and people with disabilities, and post-course student interviews. Our findings indicate that although students encountered little issues with completing the assignments, they were unlikely to retain knowledge related to accessibility. Contributions of this work include empirical findings that show that it is possible to incorporate accessibility into core courses while maintaining course integrity, but that more work needs to be done to emphasize the relationship between technical knowledge and accessibility.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451625>

¹<https://teachaccess.org/>

2 RELATED WORK

Educators have attempted to include accessibility content into their courses, often yielding satisfactory results [4, 9]. However, most researchers are instructors with an interest in accessibility, a limitation that may inhibit faculty with little knowledge in accessibility [13, 23].

2.1 Approaches to Teaching Accessibility

Research on teaching accessibility comprise four categories [1]: accessibility throughout the program, in stand-alone courses, as a theme in a course, and as independent and interchangeable modules/assignments. The first approach, including accessibility throughout a program [3, 24], introduces accessibility in fundamental technology courses. In some programs, students participated in team projects requiring consideration of a diverse group. Student projects from such research showed consideration for users with disabilities in their work [24]. Still, such programs have been discontinued due to low enrollment [3], indicating waning student interest in programmatic emphasis on accessibility.

In the second approach, stand-alone courses included accessibility as the main focus, such as on universal access [12], assistive technologies [16], and engineering accessible software [7]. Researchers using this approach report students successfully engaging accessibility and becoming more comfortable interacting with people with disabilities [12].

The third approach emphasizes accessibility as a theme in a course [2, 14, 22, 25], incorporating accessibility alongside main topics perhaps as a unifying theme, but not making up the fundamental learning objective [25]. This approach was found effective for accessibility technical skills: engineering students demonstrated a better understanding of disabilities when they were required to consider accessibility in their project [14] and were motivated to consider accessibility [2], while design students shifted perspectives toward inclusive thinking [22]. Integrating accessibility as a theme was mainly used in design and engineering courses with a Human Computer Interaction (HCI) aspect.

The fourth approach involves accessibility modules (e.g., pedagogically connected segments of lecture and related activities) created to easily integrate into other courses. Modular approaches incorporate accessibility components into existing computing [6, 17, 19] and engineering [15] courses through assignments, lectures, labs, etc. Such approaches include a variety of pedagogic and programmatic elements that align with accessibility, including focusing on different types of disabilities and accessibility solutions, and incorporating content in a mix of lectures and practical experiences [15]. One assessment of student learning through modular intervention found that accessibility awareness increased by the end of the term [17]. However, long term gains were less pronounced [5, 27].

We adopted the last approach, but incorporated modules into computing programming courses, and not in HCI focused or design courses as had been previously done. This work is similar how Waller, et. al. [24] integrated accessibility early in their program, though the impact had not been studied to understand what was needed to make them effective.

2.2 Instructors' Perspectives

Researchers investigated instructors' perceived barriers to teaching accessibility. Common frustrations expressed by instructors include lack of administrative support, student engagement, and appropriate textbook [20]. In a 2018 survey, faculty rated major barriers to teaching accessibility: that it is not a core part of the curriculum, and they do not know enough about it [23]. Further research mapping accessibility and computer science learning objectives uncovered organizational constraints and lack of personal incentive for instructors [11]. Thus, there is a need to provide materials and teaching strategies that may be easily included into existing course materials and pedagogies.

3 METHOD

In this study, we included accessibility content in programming assignments² in four course offerings: Programming in C (CS1) at Rochester Institute of Technology in Spring 2018 and Spring 2019, and Object Oriented Programming (CS2) at Creighton University in Fall 2018 and Spring 2019. The courses were taught by the second and third authors of this paper, respectively. We distributed pre- and post- surveys before and after the assignment to measure student attitudes towards accessibility and people with disabilities. We conducted follow up interviews with students to determine what they retained about accessibility.

3.1 Assignments with Added Accessibility Content

Aside from the assignments, the instructors did not explicitly teach accessibility topics in other sections of CS1 and CS2. Our aims were to remain consistent with existing teaching material (which did not already include accessibility), and to determine if the assignments alone would raise students' awareness of accessibility without disrupting the primary learning objectives.

3.1.1 CS1 and Braille Translator Assignment. CS1 was an introductory course for first-year students which covered fundamental concepts in computer science and basic coding skills. The learning objectives of CS1 include learning programming, analyzing problems, and constructing small software components. In this course offering, we created an assignment to convert ASCII text to Braille output using arrays and pointers in C. The assignment included a brief introduction on Braille, its usage, and a detailed problem breakdown, focusing on learning objectives of a: (1) good understanding of pointers and arrays in C, and (2) basic understanding of Braille as an assistive technology. Similar to other assignments in CS1, students kept an activity journal upon completing the assignment to reflect on lessons they learned and challenges they faced. On the last day of classes, the students answered a 5-point Likert-scale question about whether the Braille assignment had taught them something about accessibility. We collected the student responses to these activities for analysis.

3.1.2 CS2 and Accessibility Checkers Assignment. CS2 is a programming course focused on the object-oriented approach to software programming. Course learning objectives include Java basics,

²See Supplementary Materials for full assignments write-up

object-oriented design, and techniques to analyze and breakdown a complex problem. In line with the course topic “Files, lists and objects,” we created an Accessibility Checkers assignment to write a Java program to read a file containing accessibility test results. The assignment included a definition of an accessible web page and a test suite created by the United Kingdom’s Government Digital Services to evaluate accessibility checkers. The learning objectives of this assignment were: (1) good understanding of using Java to read from a file and to perform sequential search on a list, and (2) basic understanding of automated accessibility testing.

3.2 Data Collection and Analysis

Each graded assignment provided data on learning objectives met. Due to the nature of the CS1, which had a reflection write up for every assignment, activity journals collected reflections on the Braille assignment. Quiz questions asked CS1 students if the assignment taught them something about accessibility. Surveys collected data on student attitudes, and interviews followed up with understanding student perspectives.

3.2.1 Participants. Seventy-two students completed CS1 (Spring 2018 and 2019) and 33 students completed CS2 (Fall 2018 and Spring 2019). Of the 39 combined responses in CS1, seven were female, 31 were male and one was non-binary; mean age was 18.4 years ($SD = 0.88$). Of the 20 combined responses in CS2, four were female and 16 were male; mean age was 20.6 years ($SD = 3.05$).

3.2.2 Survey Instrument. Pre- and post- surveys measured changes in student attitudes toward accessibility and people with disabilities. Survey participation was voluntary, and students were allowed to skip any question. The survey consisted of three parts:

- Six yes/no questions about students’ experiences with people with disabilities and their perception of accessibility, e.g., how they defined accessibility, and if they had interacted with people with disabilities. Students were given space to explain their answers. “Yes” answers were scored 1 point and “no” scored 0 in total score calculation.
- Multiple choice questions on students’ knowledge of accessibility issues faced by people with specific disabilities, e.g., blindness, deafness and learning disabilities. For each disability type, students could respond that they (a) have knowledge (1 point), (b) have personal experience (2 points), or (c) don’t have knowledge.
- Questions from the Interaction with Disabled Persons (IDP) scale, a standard scale assessing 6 factors: Discomfort, Sympathy, Uncertainty, Fear, Coping and Vulnerability [8], consisting of 20 statements on a 6-point Likert scale describing emotions experienced by respondents when interacting with people with disabilities.

We qualitatively analyzed reflection assignments and open-ended survey responses to gain insights into students’ perspectives on assignments, their interactions with disabled people, and their accessibility awareness.

3.2.3 Interviews. We conducted semi-structured follow-up interviews with students after the courses ended. Guided by our survey results, we asked questions to clarify what contributed to awareness and motivation. We also asked for feedback on the courses.

We recruited students who enrolled in any of the four courses to participate in a 30-minutes interview with \$15 compensation. The interviews took place 6 months after the last course was completed. Five students were interviewed; all took CS1, we had no responses from CS2 students. In addition to clarifying questions, we asked about participants’ perception of accessibility, building accessible products, and their accessibility education.

3.2.4 Analysis. Using a deductive approach based on expected student learning outcomes about accessibility, we coded assignments, interviews and open-ended survey responses on what students reported learning and what they recalled about the accessibility assignments. Inductively, we coded how their course experiences influenced their perception of accessibility after-the-fact. Interview analysis focused on student reaction to assignment efficacy as far as expanding awareness about accessibility. We employed inductive coding on interview transcripts, and qualitatively analyzed codes for thematic findings related to our research questions. Our analysis thus focused on characterizing experiences, and understanding knowledge and awareness gained, if any.

4 FINDINGS

Assessment showed that core learning objectives were met with students successfully achieving computing learning goals as described for each course. In this section, we focus on the efficacy of our method of including accessibility.

4.1 Survey Results

Responses from students who completed both pre- and post- surveys were analyzed. We cleaned the data by removing incomplete responses, calculated total scores for the survey and used Wilcoxon Signed-Rank Test to determine if there was a significant difference between the pre- and post- scores for each factor. Answers to open ended questions were coded as part of the qualitative analysis to identify themes related to cultivating awareness and learning accessibility.

4.1.1 CS1. Thirty-two students took CS1 in Spring 2018, 40 took it in Spring 2019. Data cleaning resulted in 15 and 24 paired responses, respectively, for a total of 39, seven were female, 31 were male and one was non-binary; mean age was 18.4 years ($SD = 0.88$). We used Wilcoxon Signed-Rank Test to compare the pre- and post- survey paired responses. No significant differences were found between pre-survey answers from the 2018 and 2019 classes. We did not find a significant change in scores of the Accessibility Survey between the pre- and post- responses ($W = 521$, $Z = -0.529$, $p = 0.59$, two-tailed). We combined the two samples of IDP scale responses and compared the pre- and post- scores for each of the six factors: Discomfort, Sympathy, Uncertainty, Fear, Coping and Vulnerability. No significant change was observed for each factor.

4.1.2 CS2. Fourteen students took CS2 in Fall 2018, 19 students in Spring 2019. Data cleaning resulted in seven paired responses from 2018 and 13 from 2019, totaling 20 responses. Four were female and 16 were male, mean age was 20.6 years ($SD = 3.05$). We compared the results of pre-survey from CS2 offerings in Fall 2018 and Spring 2019 and combined the two samples since they were not significantly different. No significant change was observed between pre- and

post- results of the Accessibility Survey for most questions ($W = 72$, $Z = -0.926$, $p = 0.352$, two-tailed). However, significant difference was found between pre- and post- results on the Sympathy factor ($W = 39$, $Z = -2.25$, $p < 0.05$, two-tailed) and Fear factor ($W = 18$, $Z = -2.16$, $p < 0.05$, two-tailed). Median change in scores of the Sympathy factor (pre = 0, post = -1) and the Fear factor (pre = 0, post = -2) indicated a positive change, notably, that students perhaps felt more sympathy and less fear towards people with disabilities after the course.

4.2 Course Materials, Open-ended Survey Responses, and Interview Findings

Because survey results showed little change in awareness about accessibility, we endeavored to understand what aspects of learning may have detracted from or contributed to awareness of accessibility. Our analysis showed that students had little prior experience with disability and accessibility, and that though assignments were sometimes effective, more could be done to emphasize the relationship between accessibility and technical skill.

4.2.1 Open-ended Survey Responses. In open-ended responses, most students did not reference their accessibility assignments, with CS2 students more likely to reference their accessibility assignment. These findings indicate that most students may not have considered the assignments as key in learning about accessibility, but that some did find the assignments enlightening about accessibility.

We identified four themes in students' responses: (1) *Contexts and Relationships*- knowing a person with a disability, or having regular encounters, whether in campus clubs, through volunteering, or in team projects; (2) *Being Considerate*- including not treating people with disabilities differently. (3) *Tensions and Challenges*- including feelings of self-consciousness or discomfort around people with disabilities due to fear of being insensitive. (4) *Learning and Improving*- students sought to become more empathetic and understanding toward people with disabilities and to alleviate tensions about things they did not know. Students learned by reflecting on their personal experiences with disability:

"Especially with physical disabilities, they no longer 'stick out' to me like they used to, physical and mental disabilities often make people uncomfortable, but after being around so much physical disability - through my best friend and being around the disability community it has begun to lose the shock value associated with physical disability." (CS2 18-12)

These four themes highlighted student uncertainty around interacting with people with disabilities and their curiosity toward learning more, indicating these issues arose when students reflected on accessibility in their course work. These themes corroborate prior work documenting student reactions to learning about accessibility [14, 20, 23]. We focused interviews on how their course experiences (CS1/2 courses and other computing courses broadly) and accessibility assignments contributed to these perspectives. Our interview questions follow-up on the fourth theme identified, to understand students' journey to Learning and Improving knowledge and skills about accessibility.

4.2.2 Interview Reflections on Braille Translator Assignment. We were successful at recruiting participants only from the CS1 course, resulting in additional qualitative insight for the CS1 student perspective. In addition, the CS1 course required students to write journal reflections on most of their assignments by way of assessing personal challenges in problem solving. For the Braille assignment, we added a question about the assignment's impact on students' understanding of software engineering, to understand the effectiveness of the Braille assignment on student learning.

Most students did not specifically mention the Braille translator in their assignment reflections, but they discussed the technical difficulties encountered and commented on their learning of pointers and arrays in C. Interviews revealed that students who knew Braille did not think the assignment taught them anything new. Three students mentioned searching for additional information on Braille when doing their assignment, while one student noted the Braille assignment increased their understanding of assistive technology:

"This [helped me] understand just how unique the programs for these assistive technologies must be implemented." (CS1 18-3)

Despite survey results showing little new knowledge about accessibility per-se for CS1 students (compared with CS2 students), in interviews participants indicated increased awareness of the diversity of users after the Braille assignment:

"This helped me understand that as a software engineer, I will have to develop software that will help people, even in ways that I would not need help myself." (CS1 19-18)

Despite the pointed use of a Braille translator as the focus of the assignment, many students glossed over the application of accessibility in pursuit of learning the technical functions for pointers and arrays. This functional focus was understandable given the goals of the course; however, it was unclear why students later overlooked the secondary takeaway that Braille (an assistive device) had some relationship with technical implementation. When asked about possibly missing the accessibility focus of the Braille assignment, students acknowledged that the topic competed with the stated learning objective:

"If we were talking about how accessibility is addressed in the course, I feel it didn't help us learn anything in that...because it's mostly programming. It doesn't feel like it's highlighting accessibility." (P4)

Participants indicated that though the assignment helpfully included everything they needed to know for the Braille translator assignment, it may have prevented them from seeking more information on their own. Participants indicated that added emphasis, such as a brief discussion in class on accessibility topics, could round out the real-world implications of the domain of the assignment:

"...to help me gain a better understanding of the challenges people with disabilities are facing...it should be something that actually gives you experience of the challenges people with disabilities are facing." (P1)

4.2.3 Reflections on Accessibility in Pedagogy and Content. In contrast with the survey responses, interview participants were motivated to understand more about accessibility, and also felt that accessibility was an important technical endeavor. However, participants also described a deficiency in how well accessibility was included in their computing education. Course focus was more often on learning to build things, usually assuming a self-as-user perspective. In light of student reflections on the accessibility assignment in the CS1 course, we sought participant perspectives on including accessibility in computing courses. Overall, participants did not feel that accessibility was prioritized at all beyond awareness in their computing education.

“I don’t think there’s something I have explicitly been taught...A lot of these are to build something I learned. If I watch something, then it’s always to hear the best practice for doing something this way.” (P5)

In keeping with tech-centric notions focusing on technical skills aimed at finding solutions, most courses that students discussed (including ours) either excluded accessibility altogether, or glossed over the topic. Yet, when accessibility was covered (such as in CS1), the content was not as helpful as expected, and rarely extended beyond promoting awareness.

“Accessibility was not really highlighted in courses ... we just talk about this is something you should keep in mind when programming. Like one day explanation and then move on.” (P4)

Despite these experiences, students expressed a willingness to have in-depth discussions and practical instruction on accessibility. Participants suggested including accessibility as part of the grading rubric of student projects, or to incorporate in-depth examples of accessibility that called on more sophisticated technical problem solving. For example, P1 expressed, “It has to be normalized...I feel like the classes are limited because when we have accessibility, it’s only a separate topic and not a part of everything we do.” More specifically, participants indicated an interest in understanding how their tech-centric skills could be applied to accessibility solutions.

“...You can go more in depth about... specific examples of accessibility, explain every kind of problem... Because really, I know things need to be accessible, I don’t know how to implement it for every case.” (P4)

Ultimately, these perspectives indicate that participants were open to the idea of threading accessibility throughout courses or assignment narratives to drive home the need for technical skill and the context surrounding such a need.

5 DISCUSSION

5.1 Assessing Efficacy of the Assignments

Survey data analysis did not show a general shift in students’ attitudes towards accessibility and people with disabilities. However, survey responses for the IDP scale showed significant differences in two factors for the CS2 students: Sympathy and Fear. These results indicate students have more sympathy and less fear towards people with disabilities after taking CS2. Qualitative analysis of students’ comments revealed that students thought they learned something about accessibility through the Accessibility Checkers assignment.

Compared to the Braille assignment, the Accessibility Checkers assignment received more positive feedback from students. However, investigation of student activity journals and interviews revealed that although the Braille assignment raised accessibility awareness, students also thought the assignment did not teach them anything new. The Accessibility Checkers assignment explicitly mentioned accessibility and was directly relevant to programming, while the Braille translator assignment did not cover accessibility and involved an assistive technology (not accessibility-related programming). To address these, interview participants suggested providing more in-depth content on Braille translator and making accessibility “a focus” in the assignment.

Interviews corroborated that though some participants had prior exposure to accessibility content in other courses, they did not feel the topic was relevant to themselves and their major [5]. Student reactions to and perspectives on our accessibility assignments show that including accessibility-adjacent topics is helpful, but that practical and explicit instructions may be needed to drive home the relationship between technical skill and accessibility knowledge.

5.2 Limitations and Future Research

We were unable to recruit CS2 students to interview, and we recruited a small number of CS1 students. Thus, our findings are skewed toward the small number of students who encountered the Braille assignment, with more knowledge about how to adjust that assignment to emphasize accessibility. Our sample size is relatively small for substantive findings in our survey data. In the future, we plan on improving the design of our accessibility assignments by clearly describing the accessibility learning objective. We also plan on seeking input from computer science professors on the accessibility assignments to facilitate their adoption to the assignments.

6 CONCLUSION

Teaching accessibility in programming courses can offer computing students the needed reinforcement to the accessibility knowledge they may receive in HCI or related courses. We presented a new study on designing accessibility assignments that teaches programming and accessibility topics simultaneously. What distinguishes our work from previous “modules” is that we made accessibility assignments an original part of the course and not an add-on. Our findings across the four courses offerings showed that student learning of the core computing topics was not disrupted by including accessibility assignments and that when the accessibility topic was explicit (as in the CS2 assignment), students’ awareness of accessibility and people with disabilities increased. By creating accessibility assignments that are effective and easy to integrate in programming courses, we may decrease the gap in accessibility teaching in computing in higher education.

ACKNOWLEDGMENTS

We wish to thank Henry Luce Foundation - Clare Boothe Luce Fund for supporting this work.

REFERENCES

- [1] Catherine M. Baker, Yasmine N. El-Glaly, and Kristen Shinohara. 2020. A Systematic Analysis of Accessibility in Computing Education Research. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education* (Portland, OR, USA) (SIGCSE '20). Association for Computing Machinery, New York, NY, USA, 107–113. <https://doi.org/10.1145/3328778.3366843>
- [2] Kimberly Edginton Bigelow. 2012. Designing for Success: Developing Engineers Who Consider Universal Design Principles. *Journal of Postsecondary Education and Disability* 25, 3 (2012), 211–225.
- [3] Paul Ryan Bohman. 2012. Teaching accessibility and design-for-all in the information and communication technology curriculum: Three case studies of universities in the United States, England, and Austria. (2012).
- [4] Robert F. Cohen, Alexander V. Fairley, David Gerry, and Gustavo R. Lima. 2005. Accessibility in Introductory Computer Science. In *Proceedings of the 36th SIGCSE Technical Symposium on Computer Science Education* (St. Louis, Missouri, USA) (SIGCSE '05). Association for Computing Machinery, New York, NY, USA, 17–21. <https://doi.org/10.1145/1047344.1047367>
- [5] Paula Conn, Taylor Gotfrid, Qiwen Zhao, Rachel Celestine, Vaishnavi Mande, Kristen Shinohara, Stephanie Ludi, and Matt Huenerfauth. 2020. Understanding the motivations of final-year computing undergraduates for considering accessibility. *ACM Transactions on Computing Education (TOCE)* 20, 2 (2020), 1–22.
- [6] Y. El-Glaly, W. Shi, S. Malachowsky, Q. Yu, and D. E. Krutz. 2020. Presenting and Evaluating the Impact of Experiential Learning in Computing Accessibility Education. In *2020 IEEE/ACM 42nd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)*. 49–60.
- [7] Yasmine N El-Glaly. 2020. Teaching Accessibility to Software Engineering Students. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education*. 121–127.
- [8] Chris Forlin, Gerard Fogarty, and Annemaree Carroll. 1999. Validation of the factor structure of the Interactions with Disabled Persons Scale. *Australian Journal of Psychology* 51, 1 (1999), 50–55.
- [9] Ed Gellenbeck. 2005. Integrating Accessibility into the Computer Science Curriculum. *J. Comput. Sci. Coll.* 21, 1 (Oct. 2005), 267–273.
- [10] Yavuz Inal, Kerem Rızvanoğlu, and Yeliz Yesilada. 2019. Web accessibility in Turkey: awareness, understanding and practices of user experience professionals. *Universal Access in the Information Society* 18, 2 (2019), 387–398.
- [11] Saba Kawas, Laura Vonessen, and Amy J Ko. 2019. Teaching accessibility: A design exploration of faculty professional development at scale. In *Proceedings of the 50th ACM Technical Symposium on Computer Science Education*. 983–989.
- [12] Sri H Kurniawan, Sonia Arteaga, and Roberto Manduchi. 2010. A general education course on universal access, disability, technology and society. In *Proceedings of the 12th international ACM SIGACCESS conference on Computers and accessibility*. 11–18.
- [13] Sarah Lewthwaite and David Sloan. 2016. Exploring pedagogical culture for accessibility education in computing science. In *Proceedings of the 13th Web for All Conference*. 1–4.
- [14] Stephanie Ludi. 2007. Introducing accessibility requirements through external stakeholder utilization in an undergraduate requirements engineering course. In *29th International Conference on Software Engineering (ICSE'07)*. IEEE, 736–743.
- [15] Israel Martin-Escalona, Francisco Barcelo-Arroyo, and Enrica Zola. 2013. The introduction of a topic on accessibility in several engineering degrees. In *2013 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, 656–663.
- [16] Kerstin Matausch, Barbara Hengstberger, and Klaus Miesenberger. [n.d.]. "Assistec" – A University Course on Assistive Technologies. In *Computers Helping People with Special Needs* (2006), Klaus Miesenberger, Joachim Klaus, Wolfgang L. Zagler, and Arthur I. Karshmer (Eds.). Springer Berlin Heidelberg, 361–368.
- [17] Nidhi Rajendra Palan, Vicki L. Hanson, Matt Huenerfauth, and Stephanie Ludi. 2017. Teaching Inclusive Thinking in Undergraduate Computing. In *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility* (Baltimore, Maryland, USA) (ASSETS '17). Association for Computing Machinery, New York, NY, USA, 399–400. <https://doi.org/10.1145/3132525.3134808>
- [18] Rohan Patel, Pedro Breton, Catherine M. Baker, Yasmine N. El-Glaly, and Kristen Shinohara. 2020. Why Software is Not Accessible: Technology Professionals' Perspectives and Challenges. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3334480.3383103>
- [19] G Michael Poor, Laura M Leventhal, Julie Barnes, Duke R Hutchings, Paul Albee, and Laura Campbell. 2012. No user left behind: Including accessibility in student projects and the impact on CS students' attitudes. *ACM Transactions on Computing Education (TOCE)* 12, 2 (2012), 1–22.
- [20] Cynthia Putnam, Maria Dahman, Emma Rose, Jinghui Cheng, and Glenn Bradford. 2016. Best practices for teaching accessibility in university classrooms: cultivating awareness, understanding, and appreciation for diverse users. *ACM Transactions on Accessible Computing (TACCESS)* 8, 4 (2016), 1–26.
- [21] Anne Spencer Ross, Xiaoyi Zhang, James Fogarty, and Jacob O. Wobbrock. 2017. Epidemiology as a Framework for Large-Scale Mobile Application Accessibility Assessment. In *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility* (Baltimore, Maryland, USA) (ASSETS '17). Association for Computing Machinery, New York, NY, USA, 2–11. <https://doi.org/10.1145/3132525.3132547>
- [22] Kristen Shinohara, Cynthia L Bennett, Wanda Pratt, and Jacob O Wobbrock. 2018. Tenets for social accessibility: Towards humanizing disabled people in design. *ACM Transactions on Accessible Computing (TACCESS)* 11, 1 (2018), 1–31.
- [23] Kristen Shinohara, Saba Kawas, Andrew J Ko, and Richard E Ladner. 2018. Who teaches accessibility? A survey of US computing faculty. In *Proceedings of the 49th ACM Technical Symposium on Computer Science Education*. 197–202.
- [24] Annalu Waller, Vicki L Hanson, and David Sloan. 2009. Including accessibility within and beyond undergraduate computing courses. In *Proceedings of the 11th international ACM SIGACCESS conference on Computers and accessibility*. 155–162.
- [25] Ye Diana Wang. 2012. A holistic and pragmatic approach to teaching web accessibility in an undergraduate web design course. In *Proceedings of the 13th annual conference on Information technology education*. 55–60.
- [26] Webaim.org. 2018. Survey of Web Accessibility Practitioners #2 Results. Retrieved December 17, 2020 from <https://webaim.org/projects/practitionersurvey2/>
- [27] Qiwen Zhao, Vaishnavi Mande, Paula Conn, Sedeeq Al-khazraji, Kristen Shinohara, Stephanie Ludi, and Matt Huenerfauth. 2020. Comparison of Methods for Teaching Accessibility in University Computing Courses. In *The 22nd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, Greece) (ASSETS '20). Association for Computing Machinery, New York, NY, USA, Article 6, 12 pages. <https://doi.org/10.1145/3373625.3417013>